

Toward Verifiable Formula Extraction for Docling

A Steering Document for Dataset, Adapter, and Validation Work

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Abstract

Scientific documents contain mathematical expressions that are easy for humans to read but difficult for machines to recover faithfully. Existing document parsers and small vision-language models can already extract formulas into $\text{\LaTeX}$, but the field still lacks strong validation loops, hard real-world benchmarks, and semantic trust signals. This document frames a research and product direction: build a Docling-compatible formula extraction adapter whose first-class asset is not merely a model, but a curated dataset and validation harness that converts formula pixels into $\text{\LaTeX}$, checks renderability, measures visual equivalence, canonicalizes outputs, and selectively validates semantics with Lean where typed formalization is possible.

1 Position

The opportunity is to treat formula extraction as compilation from visual mathematical notation into structured representations. A useful system should produce $\text{\LaTeX}$, preserve document location, expose confidence and validation metadata, and support semantic validation for a subset of expressions.

Program thesis: build verifiable structured extraction from scientific documents, beginning with mathematical formulas and expanding to code, tables, charts, and proof-carrying technical knowledge.

2 Seed Context

Taru’s seed links point to Docling’s code/formula stack: CodeFormula, CodeFormulaV2, SynthFormulaNet, SynthCodeNet, ChartNet, Segment Anything, and the 2024 Vary-toy paper on reinforced vision vocabularies.

The immediate implication is that the first serious step is dataset curation and evaluation, not training a small model from scratch. CodeFormulaV2 and SmolDocling already provide strong baselines. Our edge should come from hard slices, validation, semantic structure, and integration quality.

3 Existing Baselines

Asset	Type	Relevance
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CodeFormula	Model	0.2B-class model for 120 DPI image crops of code or formulas. Outputs code or $\text{\LaTeX}$.
CodeFormulaV2	Model	0.3B-class newer baseline trained on SynthFormulaNet and SynthCodeNet. Best first baseline.
SynthFormulaNet	Dataset	6.45M image-to- $\text{\LaTeX}$ rows. Core source for sampling, validation, and benchmark construction.
SynthCodeNet	Dataset	9.33M image-to-code rows. Useful analogy for structured visual-code transcription.
SmolDocling	Model/system	256M document VLM that emits DocTags for full-page document conversion. Strategic integration anchor.
ChartNet	Dataset	1.5M chart dataset with image, code, data table, summary, and QA. Strong dataset-design template.

4 Docling Adapter Surface

Docling exposes code/formula enrichment through pipeline options. A practical integration target is:

```
from docling.datamodel.pipeline_options import CodeFormulaVlmOptions
from docling.datamodel.pipeline_options import PdfPipelineOptions

code_formula_options = CodeFormulaVlmOptions.from_preset("codeformulav2")

pipeline_options = PdfPipelineOptions(
    do_ocr=False,
    do_code_enrichment=True,
    do_formula_enrichment=True,
    code_formula_options=code_formula_options,
)
```

The adapter should return formula text into Docling’s document representation while retaining provenance, bounding boxes, confidence, validation status, and possibly alternate candidates.

5 Validation Stack

5.1 LaTeX Validation

The first validation layer checks whether predicted $\text{\LaTeX}$ compiles or renders under a controlled engine. This catches broken commands, unbalanced braces, unsupported environments, and malformed structures.

5.2 Visual Equivalence

$\text{\LaTeX}$ exact match is too brittle. Equivalent strings may render to the same expression. The validation harness should render canonical $\text{\LaTeX}$ and compare it to the original crop using image

similarity and layout-aware metrics.

5.3 Canonicalization

Canonicalization should normalize whitespace, macro aliases, bracket sizing, optional tags, and common equivalent constructs. This is an engineering asset: it improves both training data quality and evaluation stability.

5.4 Lean Validation

Lean validates typed formal mathematics, not $\text{\LaTeX}$. It should be used on a curated semantic subset where variables, domains, and assumptions can be recovered or provided.

Layer	Coverage	Trust signal
$\text{\LaTeX}$ compile	Broad	Syntactic/renderability trust
Visual match	Broad	Perceptual transcription trust
Canonicalization	Broad	Evaluation stability
Lean check	Narrow	Typed semantic trust

6 Dataset Schema

Each row should support reproducibility, validation, and future model training:

```

id
source_dataset
source_license
image_path_or_bytes
raw_latex
canonical_latex
docling_target_text
bbox_tokens
dpi
render_engine
render_status
latex_compile_status
visual_similarity_score
formula_type
complexity_tags
lean_candidate
lean_statement
lean_check_status
split_group
notes

```

7 Opportunity Matrix

Workstream		Value	Difficulty	Notes
Curated benchmark	formula	High	Medium	Best first public artifact. Shows taste and creates leverage.
LaTeX render	validator	High	Low-Medium	Required before model claims are credible.
Visual scoring	equivalence	High	Medium	Harder than string metrics; useful for robust evaluation.
Docling wrapper	adapter	High	Medium	Converts research into usable infrastructure.
CodeFormulaV2	base-line report	High	Low	Establishes known performance and failure slices.
Targeted generator	synthetic	High	Medium	Generate hard formulas after observing failures.
Lean-valid subset		Very high	High	Narrow coverage but strategically distinctive.
Fine-tuned adapter	small	Medium-High	Medium-High	Valuable only after dataset/eval exposes specific gaps.
Tool-interleaved recognizer	rec-	Medium-High	High	Inspired by DianJin-OCR-R1; use experts to cross-check predictions.
Full scientific document graph	document knowledge	Very high	Very high	Later-stage expansion after formula extraction works.

8 Strategic Bets

8.1 Bet 1: Dataset Quality Beats Another Model Demo

The field already has many model demos. A benchmark with hard, validated, domain-stratified formulas can become a durable asset.

8.2 Bet 2: Formula Extraction Needs Context

Isolated crops are useful but incomplete. Many formulas depend on page context: variable declarations, assumptions, equation numbering, captions, and surrounding prose. The adapter should eventually support both crop-only and context-aware modes.

8.3 Bet 3: Lean Is A Differentiator, Not A Universal Filter

Lean validation is powerful precisely because it is strict. Use it where the semantics can be made explicit. Keep it separate from LaTeX validation so the project does not collapse under impossible coverage expectations.

8.4 Bet 4: Compression And Reading Order Are The Frontier

The latest descendant papers around Vary-toy point toward efficient visual tokens, token pruning, optical compression, and semantic token reordering. Formula recognition is a natural testbed because mathematical layout is inherently two-dimensional.

9 Roadmap

9.1 Phase 0: Ground Truth Inspection

Sample 1,000–10,000 SynthFormulaNet rows. Manually inspect formula diversity, target formatting, location tokens, rendering consistency, and mismatch cases.

9.2 Phase 1: Validation Harness

Build a local pipeline that:

1. parses dataset rows,
2. extracts raw and canonical $\text{\LaTeX}$,
3. renders formula images,
4. records compile/render failures,
5. computes basic image similarity,
6. stratifies formula complexity.

9.3 Phase 2: Baseline Evaluation

Run CodeFormulaV2 and at least one larger general VLM baseline on a small curated split. Score exact match, normalized edit distance, compile rate, render similarity, and failure tags.

9.4 Phase 3: Hard-slice Dataset

Create a benchmark with real and synthetic examples:

- nested fractions,
- matrices and arrays,
- aligned equations,
- superscript/subscript nesting,
- integrals, sums, products,
- piecewise definitions,
- Greek/Latin ambiguity,
- noisy scanned crops,
- formulas with nearby equation numbers,
- theorem statements with assumptions.

9.5 Phase 4: Adapter Prototype

Wrap the best recognizer as a Docling-compatible formula enrichment stage with validation meta-data. If needed, LoRA/adapt on hard slices rather than the full dataset.

9.6 Phase 5: Lean Subset

Select a small formalizable subset and build a path from formula to Lean:

image $\rightarrow$ LaTeX $\rightarrow$ canonical form $\rightarrow$ typed semantic parse $\rightarrow$ Lean statement

This phase should begin with elementary algebra/calculus identities and theorem-like expressions where types and assumptions are explicit.

10 Evaluation

Metric	Scope	Purpose
Raw exact match	All examples	Simple but too strict.
Canonical exact match	All examples	Better string metric after normalization.
Token edit distance	All examples	Measures closeness of transcription.
L ^A T _E Xcompile rate	All examples	Detects malformed output.
Render similarity	All examples	Captures visual equivalence.
Complexity-slice accuracy	All examples	Reveals hidden failure modes.
Lean type-check rate	Semantic subset	Measures formal validity where applicable.
Docling integration success	PDFs	Measures practical system behavior.

11 What We Should Not Do First

- Do not train a model from scratch before building the evaluation harness.
- Do not claim Lean validates arbitrary L^AT_EX.
- Do not treat synthetic data scale as equivalent to real-world robustness.
- Do not build a generic PDF parser when Docling already provides the document pipeline.
- Do not optimize for leaderboard metrics before building failure visibility.

12 Near-term Deliverables

1. A reproducible dataset sampler for SynthFormulaNet.
2. A L^AT_EXcompile/render validator.
3. A 500–2,000 example hard benchmark.
4. A CodeFormulaV2 baseline report.

5. A failure taxonomy.
6. A Docling formula enrichment adapter prototype.
7. A small Lean-valid pilot subset.

13 Open Questions

- Which formula classes matter most for the target users: academic PDFs, textbooks, engineering docs, finance reports, or handwritten notes?
- Should the first public artifact be a benchmark, an adapter, a validation tool, or a report?
- Can visual equivalence be reliable enough without human review?
- How much context is needed to make Lean validation useful?
- Should the model emit one best $\text{L}^{\text{A}}\text{T}^{\text{E}}\text{X}$ string or multiple candidates with validator-guided reranking?

14 Source Notes

Core sources include CodeFormula, CodeFormulaV2, SynthFormulaNet, SynthCodeNet, SmolDocling, Vary-toy, GOT, DeepSeek-OCR, DeepSeek-OCR 2, DocTron-Formula, ChartNet, and the Docling code/formula enrichment documentation.

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